Supplementary tables for "Two-Phase Key Recovery of Forró: Bit Puncturing Meets Probabilistic Neutral Bits". Table numbers here are local to this document.

Table 1: Per-component PNBs of the divide-and-conquer PNB phase for Forró 5/256. The output mask of the distinguisher is split into its single-bit components Γ_i. For each component i, P_i is the set of key bits that are neutral for that component alone, listed with the key bits numbered as in equation (1) of the paper; $|P_i|$ is its size, m_i is the number of key bits that still have to be guessed for the component, and ε_i is its conditional correlation.

i	Γ_i	$\lvert P_i\rvert$	m_i	PNBs (P_i)	ε_i
1	$\Gamma_6^{(3)}[10]$	42	47	{5, 6, 7, 8, 20, 21, 22, 23, 24, 25, 59, 60, 61, 62, 63, 117, 118, 119, 120, 121, 133, 134, 135, 136, 137, 142, 143, 144, 145, 146, 147, 165, 166, 167, 168, 169, 197, 198, 199, 200, 201, 236}	0.93
2	$\Gamma_{10}^{(3)}[0]$	55	34	{16, 17, 20, 21, 22, 23, 24, 25, 59, 60, 61, 62, 63, 133, 134, 135, 136, 137, 142, 143, 144, 145, 146, 147, 165, 166, 167, 168, 169, 197, 198, 199, 200, 201, 207, 208, 209, 210, 211, 216, 217, 218, 219, 220, 221, 232, 233, 236, 237, 238, 239, 240, 241, 242, 243}	1
3	$\Gamma_{10}^{(3)}[10]$	27	62	{12, 13, 14, 15, 16, 17, 126, 127, 207, 208, 209, 210, 211, 216, 217, 218, 219, 220, 221, 228, 229, 230, 231, 232, 233, 242, 243}	0.88
4	$\Gamma_{14}^{(3)}[0]$	50	39	{16, 17, 20, 21, 22, 23, 24, 25, 59, 60, 61, 62, 63, 142, 143, 144, 145, 146, 147, 165, 166, 167, 168, 169, 197, 198, 199, 200, 201, 207, 208, 209, 210, 211, 216, 217, 218, 219, 220, 221, 232, 233, 236, 237, 238, 239, 240, 241, 242, 243}	1

Table 2: Trail enumeration for f_1 in the 5.5-round attack. The enumeration starts from the target bit $v_{14}^{[18]}[27]$ and goes forward one operation at a time. In each row, the mask on the left of the arrow is replaced by masks on the inputs of the operation that produced it: a modular addition [add], an XOR [xor], or a renaming [rename] of a word that the subround leaves unchanged. *Red italic* bits are active in every kept trail and green bits only in some of them. The column #coeffs gives the number of Walsh coefficients kept after the step, and ρ^2 the puncturing correlation reached so far. The last row is the final key addition: each remaining state word is expressed through the keystream word $\bar{z}_j$ and the key word k_j; its entries give the number of coefficients, bounded by two to the number of keystream and key bits kept, and the final correlation ρ_1^2.

Subround	Active	#coeffs	ρ^2
18	$v_{14}^{[18]}[27]$	1	1
19	$v_{14}^{[18]}[27] \leftarrow \{t_{14}[27, 26{-}19],\ t_1[27, 26{-}19]\}$ [add]	510	$2^{-0.01}$
	$t_1[27, 26{-}19] \leftarrow \{v_1^{[19]}[27, 26{-}19],\ t_2[27, 26{-}19]\}$ [xor]	510	$2^{-0.01}$
	$t_{14}[27, 26{-}19] \leftarrow \{v_{14}^{[19]}[22, 21{-}17],\ v_1^{[19]}[26{-}19]\}$ [add]	31048	$2^{-0.03}$
	$t_2[27, 26{-}19] \leftarrow \{v_2^{[19]}[31{-}29, 3, 2{-}0],\ v_6^{[19]}[27, 26{-}20]\}$ [add]	317376	$2^{-0.09}$
20	$v_2^{[19]}[31{-}29, 3, 2{-}0] \leftarrow \{v_2^{[20]}[31{-}29, 3, 2{-}0],\ t_3[31{-}29, 3, 2{-}0]\}$ [xor]	317376	$2^{-0.09}$
	$t_3[31{-}29, 3, 2{-}0] \leftarrow \{v_3^{[20]}[11, 10{-}5],\ v_7^{[20]}[31, 30, 3, 2, 1]\}$ [add]	357416	$2^{-0.76}$
21	$v_3^{[20]}[11, 10{-}5] \leftarrow \{v_3^{[21]}[11, 10{-}5],\ t_0[11, 10{-}5]\}$ [xor]	357416	$2^{-0.76}$
	$t_0[11, 10{-}5] \leftarrow \{v_0^{[21]}[19, 18{-}13],\ v_5^{[21]}[11, 10{-}8]\}$ [add]	1288208	$2^{-1.13}$
22	$v_6^{[21]}[27, 26{-}20] \leftarrow \{t_6[5, 4{-}2],\ t_{11}[27, 26{-}21]\}$ [add]	2910336	$2^{-1.22}$
	$v_1^{[21]}[26{-}19] \leftarrow \{t_1[26{-}21],\ t_6[26{-}24, 5, 4{-}2]\}$ [add]	4511424	$2^{-1.43}$
	$v_0^{[21]}[19, 18{-}13] \leftarrow \{v_0^{[22]}[19, 18{-}13],\ t_1[26{-}21, 19, 18{-}13]\}$ [xor]	4511424	$2^{-1.43}$
	$t_{11}[27, 26{-}21] \leftarrow \{v_{11}^{[22]}[27, 26{-}21],\ v_{12}^{[22]}[27, 26{-}24]\}$ [add]	8384832	$2^{-1.53}$
	$t_6[26{-}24, 5, 4{-}2] \leftarrow \{v_6^{[22]}[26{-}24, 5, 4{-}2],\ v_{11}^{[22]}[27, 26{-}21, 5, 4{-}2]\}$ [xor]	8384832	$2^{-1.53}$
	$t_1[26{-}21, 19, 18{-}13] \leftarrow \{v_1^{[22]}[31{-}29, 27, 26{-}21, 2{-}0],$ $v_6^{[22]}[25{-}23, 19, 18{-}16, 5, 4{-}2]\}$ [add]	20479104	$2^{-2.13}$
	$v_{14}^{[22]}[22, 21{-}17] = v_{14}^{[19]},\ v_5^{[22]}[11, 10{-}8] = v_5^{[21]},\ v_3^{[22]}[11, 10{-}7] = v_3^{[21]}$ [rename]	20479104	$2^{-2.13}$
end	$v_{14}^{[22]}[22, 21{-}17],\ v_{12}^{[22]}[27, 26{-}24],\ v_7^{[22]}[31, 30, 3, 2, 1],$ $v_6^{[22]}[25{-}23, 19, 18{-}16, 5, 4{-}2],\ v_5^{[22]}[11, 10{-}8],$ $v_{11}^{[22]}[27, 26{-}21, 5, 4{-}2] \leftarrow \{\bar{z}_{11}[27, 26{-}23, 5, 4, 3],\ k_{11}[27, 26{-}21, 5, 4{-}0]\},$ $v_3^{[22]}[11, 10{-}7] \leftarrow \{\bar{z}_3[11, 10{-}8],\ k_3[11, 10{-}6]\},$ $v_2^{[22]}[31, 3, 2{-}0] \leftarrow \{\bar{z}_2[31{-}28, 3, 2{-}0],\ k_2[31{-}29, 3, 2{-}0]\},$ $v_1^{[22]}[31{-}29, 27, 26{-}21, 2{-}0] \leftarrow \{\bar{z}_1[31{-}28, 27, 26{-}23, 2{-}0],\ k_1[31{-}28, 27, 26{-}19, 2{-}0]\},$ $v_0^{[22]}[19, 18{-}13] \leftarrow \{\bar{z}_0[19, 18{-}15],\ k_0[19, 18{-}12]\}$	$\leqslant 2^{55+44}$	$2^{-2.96}$

Table 3: Trail enumeration for f_2 in the 5.5-round attack. The enumeration starts from the target bit $v_0^{[21]}[8]$ and goes forward one operation at a time. In each row, the mask on the left of the arrow is replaced by masks on the inputs of the operation that produced it: a modular addition [add], an XOR [xor], or a renaming [rename] of a word that the subround leaves unchanged. *Red italic* bits are active in every kept trail and green bits only in some of them. The column #coeffs gives the number of Walsh coefficients kept after the step, and ρ^2 the puncturing correlation reached so far. The last row is the final key addition: each remaining state word is expressed through the keystream word $\bar{z}_j$ and the key word k_j; its entries give the number of coefficients, bounded by two to the number of keystream and key bits kept, and the final correlation ρ_2^2.

Subround	Active	#coeffs	ρ^2
21	$v_0^{[21]}[8]$	1	1
22	$v_0^{[21]}[8] \leftarrow \{v_0^{[22]}[8],\ t_1[8]\}$ [xor]	1	1
	$t_1[8] \leftarrow \{v_1^{[22]}[16, 15{-}8],\ v_6^{[22]}[8, 7{-}0]\}$ [add]	766	1
end	$v_6^{[22]}[8, 7{-}0]$, $v_1^{[22]}[16, 15{-}8] \leftarrow \{\bar{z}_1[16, 15{-}6],\ k_1[16, 15{-}9]\}$, $v_0^{[22]}[8] \leftarrow \{\bar{z}_0[8, 7{-}0],\ k_0[8, 7{-}0]\}$	$\leqslant 2^{26+15}$	$2^{-0.01}$

Table 4: Trail enumeration for f_3 in the 5.5-round attack. The enumeration starts from the target bit $v_{11}^{[22]}[10]$ and goes forward one operation at a time. In each row, the mask on the left of the arrow is replaced by masks on the inputs of the operation that produced it: a modular addition [add], an XOR [xor], or a renaming [rename] of a word that the subround leaves unchanged. *Red italic* bits are active in every kept trail and green bits only in some of them. The column #coeffs gives the number of Walsh coefficients kept after the step, and ρ^2 the puncturing correlation reached so far. The last row is the final key addition: each remaining state word is expressed through the keystream word $\bar{z}_j$ and the key word k_j; its entries give the number of coefficients, bounded by two to the number of keystream and key bits kept, and the final correlation ρ_3^2.

Subround	Active	#coeffs	ρ^2
22	$v_{11}^{[22]}[10]$	1	1
end	$v_{11}^{[22]}[10] \leftarrow \{\bar{z}_{11}[10, 9{-}0],\ k_{11}[10, 9{-}0]\}$	3070	1